

## SCH4U Unit Test 3

| K/U | A | T  | C  | Total |
|-----|---|----|----|-------|
| 51  | 8 | 11 | 10 | 80    |
|     |   |    |    |       |

### Multiple Choice

*Identify the letter of the choice that best completes the statement or answers the question*

- In an exothermic reaction, the temperature of the surroundings increases because the molecules in the surrounding have [3]
  - A lower kinetic energy
  - A lower potential energy
  - A greater kinetic energy
  - A greater potential energy
  - Not changed in either kinetic or potential energy
- The molar heat of vaporization of water is 42kJ/mol. How much energy is released by the condensation of 3.0g of water?[3]
  - 0.88kJ
  - 7.0kJ
  - 130kJ
  - 250kJ
  - 0.07kJ
- Which of the following reactions is exothermic?[3]
  - $\text{H}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g}) \Delta\text{H} = -243\text{kJ/mol}$
  - $1/2 \text{N}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) + 90.7\text{kJ} \rightarrow \text{NO}(\text{g})$
  - $\text{CO}(\text{g}) + 111\text{kJ} \rightarrow \text{C}(\text{s}) + 1/2 \text{O}_2(\text{g})$
  - $1/2 \text{H}_2(\text{g}) + 1/2 \text{I}_2(\text{g}) \rightarrow \text{HI}(\text{g}) \Delta\text{H} = +26\text{kJ/mol}$
  - $\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) - \Delta\text{H} = -243\text{kJ/mol}$
- Consider the following equation for the combustion of hydrogen:  
 $\text{H}_2(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{g}) + 243\text{kJ}$   
In order to produce 1215kJ of heat, how many grams of  $\text{H}_2$  must burn?[3]
  - 12.0g
  - 0.250g

- c. 0.100g
  - d. 8.00g
  - e. 10.0g
5. What is the average rate of production of  $\text{H}_2(\text{g})$  in the reaction of zinc with hydrochloric acid if after 30s the volume of hydrogen is 20mL, and after 1 minute, the volume is 62mL?[3]
- a. 62mL/s
  - b. 1.4mL/s
  - c. 62mL/min
  - d. either a or b
  - e. either b or c
6. Generally, temperature affects the rate of a reaction in which of the following ways?[3]
- a. Increasing the temperature reduces the rate of the reaction
  - b. Decreasing the temperature decreases the rate of the reaction
  - c. Increasing the temperature increases the rate of the reaction
  - d. Both a and b are correct
  - e. Both b and c are correct
7. Using the standard heats of formation, what is the enthalpy change per mole of  $\text{CO}_2$  for the following reaction?[3]
- $$\text{CO}(\text{g}) + 1/2 \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$$
- a. -506kJ
  - b. -284kJ
  - c. +284kJ
  - d. +506kJ
  - e. -617kJ
8. Based on the heat of formation of  $\text{MgO}$ , how much energy is required to decompose 48.6g of  $\text{MgO}$ ?[3]
- a. 603kJ
  - b. 725kJ
  - c. 301kJ
  - d. 60.3kJ
  - e. 1504kJ
9. A catalyst is only effective if it[3]
- a. Is present in large quantities
  - b. Is added before the reactants come in contact with each other
  - c. Is specific to the particular reaction
  - d. Is heated before it is added to the reaction
  - e. All of the above

10. If for the reaction,  $aX + bY \rightarrow \text{products}$ , the rate law is determined to be  $r = k[X]^1[Y]^2$ , then the order of the reaction is which of the following?[3]

- a. 1
- b. 2
- c. 3
- d. 4
- e. 5
- f.

11. The half-life of a substance is[3]

- a. Half the time for the reactant to be completely consumed
- b. The time for the reaction to be used up
- c. The time for all of the reactant to be used up
- d. The time for half the reactant to be used up
- e. None of the above

12. Which of the following is not part of the collision theory?[3]

- a. A chemical system consists of particles that are in constant motion
- b. A reaction must involve collisions of particles
- c. An effective collision has sufficient energy and correct orientation
- d. An ineffective collision can still cause a reaction
- e. The rate of the reaction depends on the frequency of effective collisions

13. The activated complex[3]

- a. is an unstable molecule
- b. has the maximum potential energy possible
- c. may continue on to produce products
- d. may revert to reactants
- e. all of the above

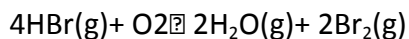
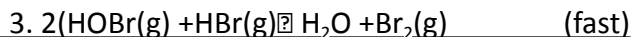
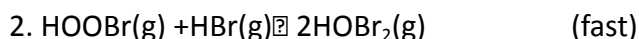
14. When a reaction is broken down into a number of elementary steps, this is known as a [3]

- |                         |                      |
|-------------------------|----------------------|
| a. reaction progression | d. reaction step     |
| b. reaction mechanism   | e. none of the above |
| c. reaction             |                      |

15. Reaction intermediates are [3]

- a. steps within a reaction mechanism
- b. the step that determines the rate of the reaction
- c. products formed in the overall equation
- d. products formed within some steps of the reaction mechanism

e. none of the above



16. Consider the above reaction mechanism. The rate-determine step of this reaction is [3]

a. elementary step 1

b. elementary step 2

c. elementary step 3

d. elementary steps 2 and 3

e. impossible to tell from this information

17. The theoretical effect of an increase in the initial concentration of a reactant can be explained in terms of collision theory because it affects [3]

I. the collision geometry involved in the reaction

II. the total number of collisions that occur

III. the fraction of collisions that are effective

IV. the required activation energy for a reaction

a. Both I and IV

b. I, II and III are true

c. Both II and III

d. Both III and IV

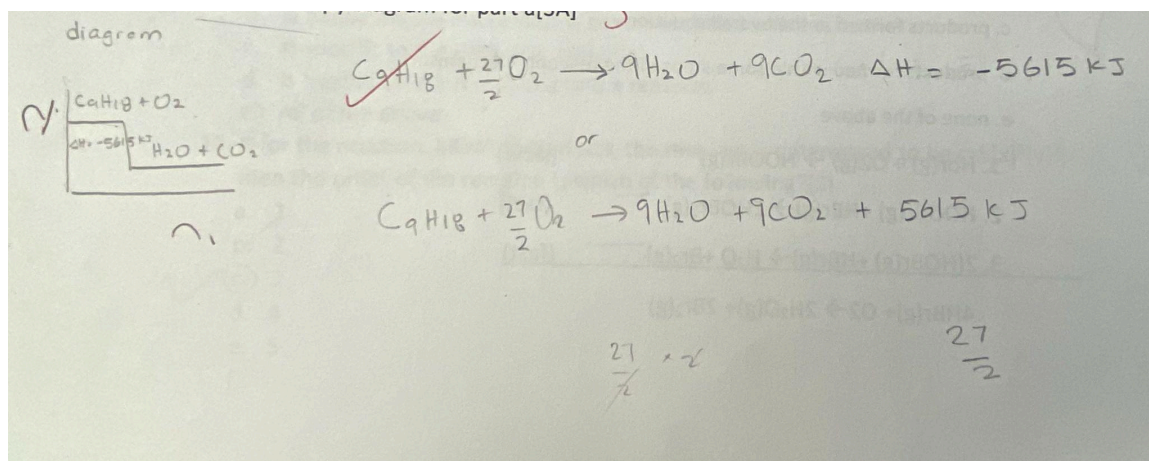
e. II only

### Short Answer

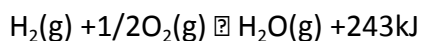
18.  $\Delta H_{\text{combustion}}$  for  $\text{C}_9\text{H}_{18} = -5615\text{kJ}$

a. write the thermochemical equation for the combustion of  $\text{C}_9\text{H}_{18}$  [3T]

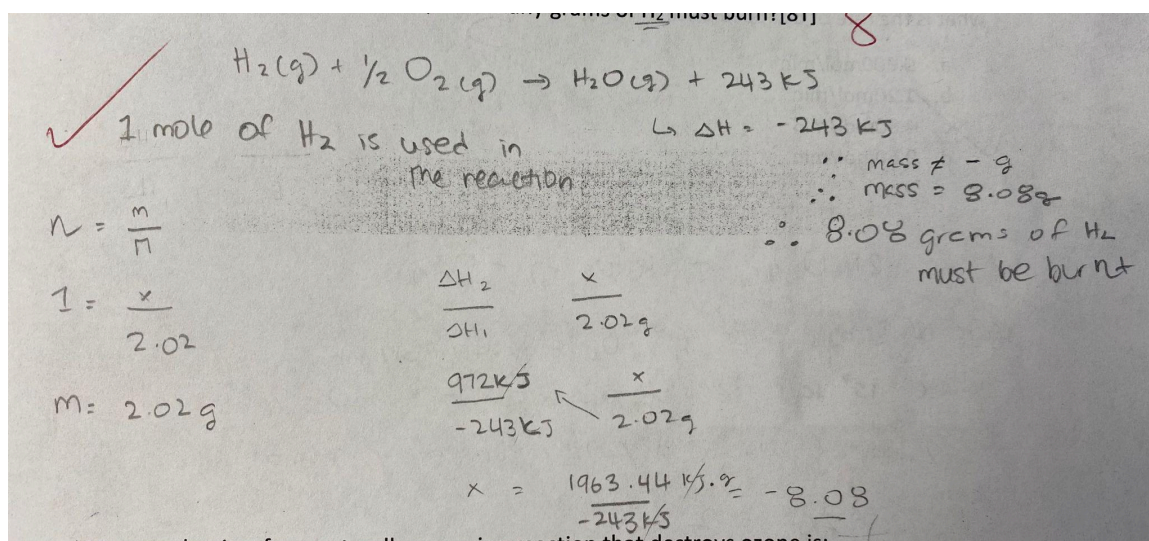
b. Draw Enthalpy diagram for part a [3A]



19. Consider the following equation for the combustion of  $H_2$  (hydrogen):



In order to produce 972 kJ of heat, how many grams of  $H_2$  must burn? [8T]

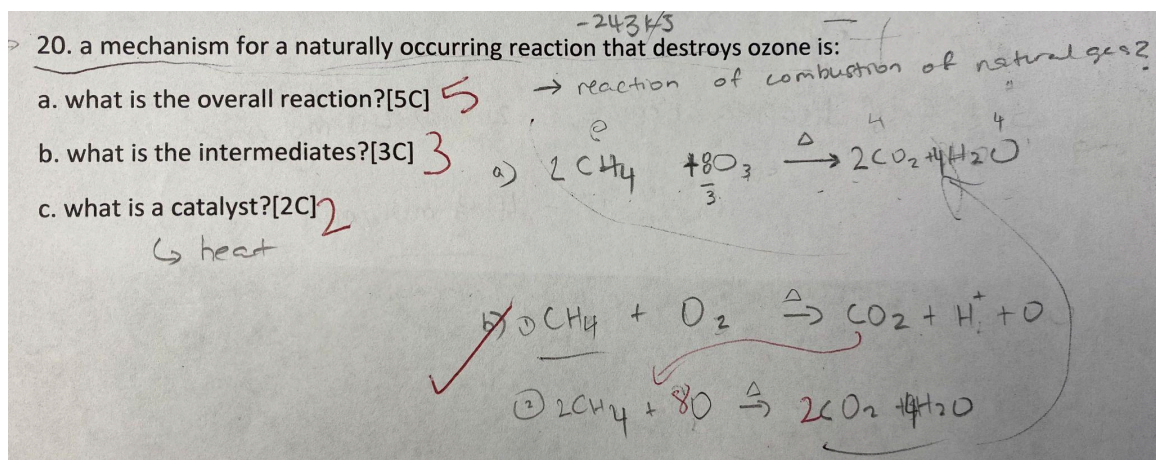


20. a mechanism for a naturally occurring reaction that destroys ozone is:

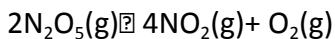
a. what is the overall reaction? [5C]

b. what is the intermediates?[3C]

c. what is a catalyst?[2C]



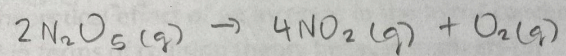
21. The decomposition of dinitrogen pentoxide is described by the reaction:



If the rate of appearance of  $\text{O}_2$  is equal to  $2.40\text{mol/min}$ , what is the rate of disappearance of  $\text{N}_2\text{O}_5$ [5A]

- $0.600\text{mol/min}$
- $1.20\text{mol/min}$
- $4.80\text{mol/min}$
- $9.60\text{mol/min}$





rate of disappearance of  $\text{O}_2 = 2.40 \text{ mol/min}$

what is rate of disappearance of  $\text{N}_2\text{O}_5$ ?

2 mols of  $\text{N}_2\text{O}_5$  disappear for every 1 mol of  $\text{O}_2$  formed

$$\therefore \frac{2}{1} = 2$$

$$\text{rate of disappearance} = 2 \times 2.40 \text{ mol/min}$$

$$= \underline{\underline{4.8 \text{ mol/min}}}$$